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Multifrequency Dielectric and Microscopic Analysis of Water in Oil and Oil in Water Emulsions

N. A. Albenayyan, Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia; A. M. Hassan, Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia / Department of Petroleum Engineering, King Fahd University of Petroleum & Minerals, Saudi Arabia; A. H. El-Husseiny, Geosciences Department, King Fahd University of Petroleum & Minerals, Saudi Arabia / Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia; M. A. Mahmoud, Department of Petroleum Engineering, King Fahd University of Petroleum & Minerals, Saudi Arabia / Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia

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Abstract

Emulsions are common in the oil industry, affecting production, processing, and treatment strategies due to their complex phase behavior and stability challenges. Various techniques are employed to identify emulsion types and evaluate their stability characteristics. This study examines the dielectric and microscopic properties of water-in-oil (W/O) and oil-in-water (O/W) emulsions across a wide frequency range (4 MHz to 3 GHz), focusing on permittivity, conductivity, and structural characteristics, seeking to distinguish between oil-based, water-based emulsions and bulk oil to determine effective treatment decisions. Dielectric measurements and imaging techniques, such as microscopy, were utilized to investigate droplet size, distribution, and phase boundaries, providing comprehensive insights into emulsion behavior. Two formulations were tested: O/W emulsions containing 50% crude oil and 50% water; W/O emulsions containing 50% water and 50% crude oil. The samples were prepared via controlled dropwise mixing at 3000 rpm for 20 minutes without surfactants, allowing natural droplet formation and enabling consistent comparison of dielectric and morphological properties across emulsion types. The results revealed challenges in stabilizing emulsions due to density differences, internal-to-external phase ratios, and mixing conditions, with instability particularly evident at low initial mixing speeds (990 rpm). The dielectric analysis revealed distinct behaviors between W/O and O/W emulsions. Dielectric measurements were performed for W/O and O/W emulsions over two days. On day one, W/O emulsions showed higher initial permittivity (~ 84) at 4MHz, decreasing with frequency, while O/W values were slightly lower (~ 80) at the same frequency range. By day two, the permittivity values had dropped across all samples, suggesting structural changes. However, the conductivity was increased with frequency in both types. Interestingly, W/O emulsions exhibited higher conductivity on day one (~ 0.24 S/cm) compared to O/W (~ 0.19 S/cm), possibly due to connected water droplets in the W/O forming conductive paths. On day two, O/W emulsions

showed higher conductivity (~ 0.23 S/cm), aligning with the expected behavior of continuous water phases. Across both days, the permittivity was decreased while the conductivity was increased with frequency, indicating polarization and dispersion effects due to the presence of multiple interfaces within the emulsion samples. These trends reflect the role of internal structure, droplet distribution, and phase continuity in determining dielectric behavior. The results emphasize how emulsion type and time impact electrical properties, which is essential for understanding emulsion stability and performance in practical applications. The distribution within the fluid of the homogeneous emulsion, observed under microscope magnifications ranging from 20 to 200 μm , shows that the dielectric result represents an average of the entire system. Overall, the dielectric method offers a reliable and non-invasive approach for distinguishing emulsion types and assessing their structural stability over time, providing critical insights that support effective treatment strategies in petroleum applications.

Keywords: Emulsion Stability, Conductivity, Permittivity, Dielectric Approach, Microscopy

Introduction

Emulsions play a critical role in various industrial processes, particularly in enhanced oil recovery (EOR), drilling fluids, and reservoir characterization. Water and oil emulsions influence multiphase flow behavior, wettability alteration, and oil mobility under subsurface conditions (Kokal, 2005; Al-Anssari et al., 2021). Understanding the fluid behavior in porous media, this study investigates the formation, stability, and microstructural features of W/O emulsions under reservoir-representative conditions. The experimental work used crude oil and synthetic brine to simulate the reservoir environment, with particular focus on salinity, ionic strength, and mixing intensity. Optical microscopy and image-based analysis were employed to characterize droplet size distribution and coalescence behavior, offering practical insights relevant to flow assurance, emulsion treatment strategies, and chemical design in petroleum systems (Olajire, 2014; Czarnecka et al., 2022).

Water-in-oil (W/O) and oil-in-water (O/W) emulsions are complex multiphase systems frequently encountered in petroleum production, transport, and processing operations. Their stability and microstructure are critical parameters influencing flow behavior, separation efficiency, and the interpretation of reservoir fluid properties. Understanding the formation mechanisms and dielectric behavior of emulsions is essential for improved fluid characterization and process optimization, especially in enhanced oil recovery (EOR) and dielectric logging applications (Schramm, 1992; Kokal, 2005). The dielectric properties of emulsions provide insights into their internal composition, phase distribution, and interfacial polarization. These properties are governed by multiple factors including droplet size, salinity, water cut, and the presence of natural surfactants or solids (Asami, 2002; Sheng, 2011). When subjected to electromagnetic fields, emulsions exhibit relaxation phenomena due to interfacial charge accumulation, commonly modeled using Maxwell–Wagner–Sillars (MWS) theory (Maxwell, 1873; Wagner, 1914; Sillars, 1937). These phenomena can be quantified using dielectric spectroscopy, offering a non-invasive technique to investigate the microstructure and composition of emulsions across a wide frequency range. In this study, water-in-oil emulsions were prepared under controlled conditions and examined using optical microscopy and broadband dielectric spectroscopy. The aim was to correlate microscopic features, such as droplet morphology and size distribution, with the emulsions' dielectric response. This combined approach provides a more comprehensive understanding of emulsion behavior, particularly in terms of dielectric isolation and its implications for petrophysical analysis and fluid interpretation in oilfield environments.

Emulsions are thermodynamically unstable dispersions of two immiscible liquids, typically water and oil, where one phase is dispersed in the other in the form of fine droplets. In petroleum systems, the most common types are water-in-oil (W/O) and oil-in-water (O/W) emulsions, depending on the continuous phase. A third, less stable type known as multiple emulsions (e.g., W/O/W) may also form under specific conditions involving surfactants and temperature gradients (Schramm, 1992; Kokal, 2005). The presence

and type of emulsion significantly affect both electrical conductivity and permeability in porous media. In W/O emulsions, water droplets are isolated within an insulating oil phase, reducing the overall electrical conductivity despite high water content (Ghosh & Sen, 1991). Conversely, in O/W systems, continuous water networks facilitate ionic conduction, resulting in higher dielectric constants and effective conductivity (Asami, 2002). This distinction is crucial when interpreting dielectric logs or resistivity measurements in oil-bearing formations.

Moreover, emulsions can obstruct pore throats, reducing permeability by forming internal blockages or interfacial films within porous rocks (Kokal, 2002). This restriction of flow can lead to significant formation damage, particularly in low-permeability or tight formations where minor pore blockage has a pronounced impact on fluid mobility. The stability and viscosity of emulsions are influenced by a range of factors, including droplet size, salinity, oil–water ratio, temperature, and the presence of natural surfactants or emulsifiers such as asphaltenes and resins. These components can stabilize emulsions by adsorbing at the oil–water interface, forming rigid interfacial films that resist coalescence. As a result, emulsions may persist under reservoir or laboratory conditions, mimicking or even exacerbating formation damage. This behavior poses challenges during core flooding experiments, where residual emulsions can skew permeability measurements or mask the actual displacement efficiency (Abdallah et al., 2007). Therefore, detailed characterization of emulsion type, internal structure, and stability is crucial not only for accurate dielectric modeling but also for predicting and managing multiphase flow behavior in complex, heterogeneous reservoir environments.

Dielectric properties describe the ability of a material to store and dissipate electric energy when subjected to an external electromagnetic field. Specifically, the complex dielectric constant—composed of a real part (ϵ'), representing permittivity, and an imaginary part (ϵ''), representing dielectric loss—captures how materials respond to alternating electric fields across various frequencies. In petroleum engineering, these properties have become increasingly important for evaluating reservoir fluids, identifying hydrocarbon zones, and monitoring enhanced oil recovery (EOR) processes (Asami, 2002; Knight & Endres, 2005). Dielectric behavior is sensitive to fluid composition, salinity, saturation, and temperature, making it a valuable diagnostic in both laboratory and field-scale measurements. For instance, formation dielectric logs are widely used in wireline and logging-while-drilling (LWD) tools to estimate water saturation and hydrocarbon type. Since water has a high dielectric constant (~ 80) compared to oil ($\sim 2\text{--}5$), variations in permittivity within a formation can be used to distinguish between water-bearing and hydrocarbon-bearing zones (Clavier et al., 1984). Additionally, frequency-dependent dielectric measurements can reveal the presence of bound water, clay content, and emulsion structures in complex reservoirs. In the context of emulsions, dielectric spectroscopy offers a non-invasive method to characterize internal microstructure, particularly the distribution of water droplets in water-in-oil systems. These emulsions influence reservoir flow behavior and may form during water flooding or polymer injection. By analyzing dielectric response over a range of frequencies, engineers can assess droplet connectivity, coalescence, and emulsion stability—factors critical for designing treatment strategies and improving separation efficiency (Ghosh & Sen, 1991; Shumskayte et al., 2022). Beyond fluids, dielectric methods are applied in digital rock physics, where dielectric models of porous rocks are used to simulate electrical response and correlate with porosity, permeability, and fluid type. Studies combining dielectric data with NMR, CT imaging, or microscopy have shown strong potential in estimating irreducible water saturation and identifying fluid-rock interactions at the pore scale (Wong et al., 2019; Melkozerova et al., 2022). As petroleum reservoirs become more heterogeneous and complex, especially in tight formations and unconventional plays, the role of dielectric properties is expected to grow. Their integration with multi-physics tools and machine learning models is opening new pathways for non-destructive reservoir characterization with greater resolution and reliability.

Although several techniques have been applied to investigate emulsion stability, the use of dielectric spectroscopy remains relatively underexplored, particularly for time-dependent analysis. In this study, a multi-frequency dielectric approach was adopted to characterize the dynamic behavior of water-in-

oil emulsions and monitor changes in their dielectric properties over time. This technique enables the identification of interfacial polarization effects and the evolution of electrical conductivity and permittivity as a function of emulsion aging. In parallel, optical microscopy was employed to capture high-resolution images of the emulsion microstructure. These visual observations supported the dielectric data by providing insights into droplet size distribution, coalescence, and overall phase behavior. Together, these complementary methods provide a comprehensive understanding of emulsion stability across different types and over time.

Materials and Methods

This study is conducted to investigate the microstructural and dielectric characteristics of water-in-oil and oil-in-water emulsions. The experimental workflow includes sample preparation, optical microscopy imaging, and dielectric property measurement using a dielectric spectroscopy system. The emulsion system was prepared using a crude oil sample provided by a local petroleum field and deionized water. No surfactants were added to isolate the behavior of natural emulsifying agents present in the crude oil. The oil and water were mixed in defined volume ratios to simulate field-relevant emulsion compositions. Brine samples of 3 wt.% KCl was also used to study the effect of ionic content on the dielectric properties.

The emulsion samples were manually prepared using a high shear digital lab mixer to ensure uniform dispersion of water droplets within the oil and water phases. The mixing time and speed were kept constant for all samples to maintain reproducibility. Microscope images were taken and dielectric properties were measured immediately after mixing to ensure that the phases did not separate from each other and to enable more consistent imaging and measurement. To characterize the morphology of the emulsions, small aliquots of each sample were placed onto clean glass slides. An optical microscope was used to capture high-resolution images under bright field illumination at 20, 50, and 200 μm magnification. Image analysis was conducted to evaluate droplet size distribution, dispersion quality, and coalescence tendencies. The imaging results allowed correlation with dielectric measurements.

The dielectric properties of the emulsions were measured using a broadband dielectric spectroscopy system (Dielectric Assessment Kit) as shown in [Figure 1](#). High-precision measurements of dielectric parameters (permittivity and conductivity) were obtained over a broad frequency range. The system measures the complex permittivity (both real and imaginary components) over a frequency range spanning from 4 MHz to 3 GHz. Emulsion samples were loaded into measurement beakers designed for liquid or semi-liquid materials, and temperature was maintained at ambient laboratory conditions throughout the tests. Before measurement, the system was calibrated using standard materials (air, short, and deionized water) according to the manufacturer's instructions. For each sample, multiple measurements were taken to ensure repeatability. The dielectric data were processed and plotted to assess relaxation behavior, interfacial polarization, and any shifts in permittivity caused by varying water content or salinity. The dielectric response was then interpreted in the context of an emulsion structure observed under the microscope. The microscopy and dielectric tests were carried out on the same emulsion batches to ensure direct correlation between structure and electrical behavior. Equipment was cleaned thoroughly between runs to avoid cross-contamination, and all measurements were conducted under controlled laboratory conditions to minimize variability by utilizing the tools shown in [Figure 2](#).

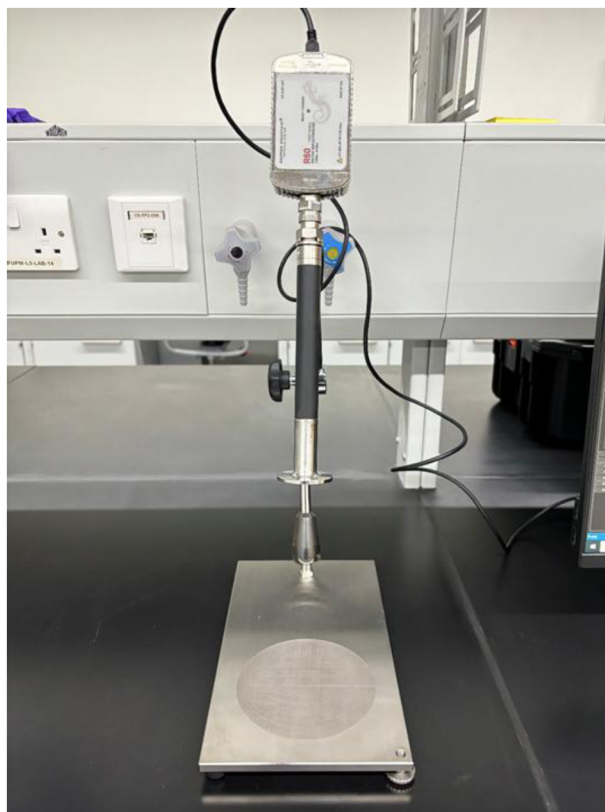


Figure 1—Multi-frequency dielectric system for non-destructive monitoring of emulsion stability over time.

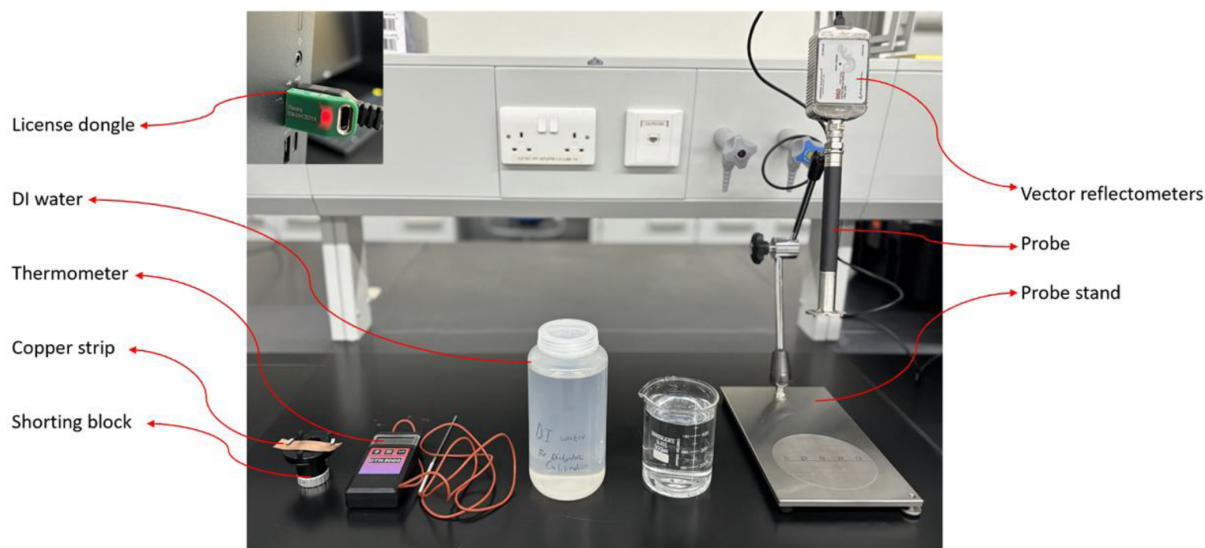


Figure 2—Dielectric measurement setup with calibration tools for accurate and repeatable sample testing.

The experimental workflow begins with the preparation of emulsion media samples under controlled conditions as presented in Figure 3. A small portion of each sample is extracted and examined under an optical microscope to evaluate droplet size and distribution. Following this, the dielectric measurement device is calibrated to ensure accurate data acquisition. The prepared emulsion sample is then introduced into the dielectric probe, where broadband dielectric spectroscopy is conducted. From the acquired data, permittivity and conductivity values are extracted and plotted as a function of frequency. The results are

then validated for consistency and finally compared across different emulsion conditions to identify trends related to stability.

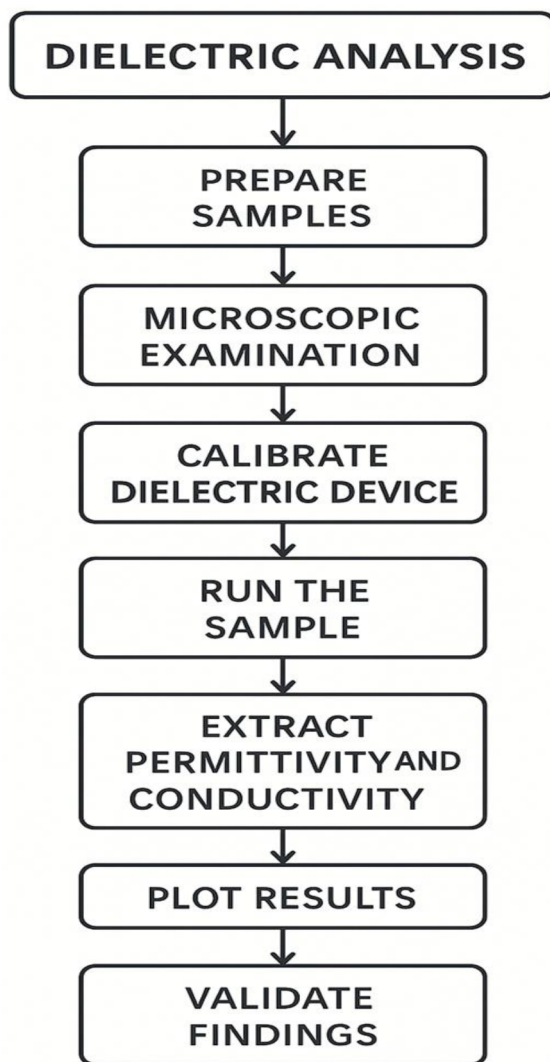


Figure 3—Experimental workflow for combined dielectric and microscopic characterization to correlate emulsion structure and stability.

A multi-frequency dielectric assessment system was used in this work to study the emulsion stability over time. The system in [Figure 1](#) measures both permittivity and conductivity across a wide frequency range, enabling detection of structural changes in emulsions over time. Its non-destructive nature allows repeated measurements on the same sample for accurate monitoring of stability behavior. The dielectric measurement setup and calibration tools are below in [Figure 2](#). The image shows the full setup used for dielectric measurements, including the vector reflectometer, coaxial probe, and supporting calibration tools. DI water, copper strip, and shorting block were used to calibrate the system before sample testing to ensure accuracy and repeatability.

The experimental workflow for dielectric and microscopic characterization of emulsions, as illustrated in [Figure 3](#), presents the sequential steps followed throughout the study. This includes sample preparation, microscopic examination, dielectric measurements, and result interpretation. The integrated methodology enables accurate correlation between the emulsion's microstructural features and its dielectric behavior over time.

Results and Discussion

This work seeks to explore the relationship between the internal architecture of water-in-oil emulsions, including droplet morphology, spatial distribution, and the extent of phase merging, and their dielectric performance across a wide frequency spectrum. By combining dielectric techniques with microscopic visualization on identical samples, the study provides a multidimensional perspective on how emulsion structure influences its electrical signature. These findings contribute to more accurate fluid classification and improved interpretation of dielectric logs in subsurface environments. Additionally, the approach offers a foundation for integrating realistic dielectric descriptors into advanced reservoir modeling and digital rock analysis workflows.

Dielectric Response

The permittivity profiles of the emulsions were measured and are presented in Figure 4. These profiles display the variation of permittivity as a function of frequency on the first day for both water-in-oil (W/O) and oil-in-water (O/W) emulsions. To ensure accuracy and reproducibility, three separate measurements were performed for each emulsion type, helping to minimize experimental variability and account for the inherent complexity of emulsion systems. The recorded permittivity values ranged from approximately 84 at 4 MHz to 3.2 at 3000 MHz, reflecting the expected dispersion behavior over a wide frequency spectrum. Notably, the O/W emulsions exhibited slightly lower permittivity values compared to W/O emulsions, which may be attributed to differences in phase continuity and droplet arrangement. Across all samples, a consistent decreasing trend in permittivity with increasing frequency was observed, suggesting significant interfacial polarization (Maxwell–Wagner effects) occurring at the oil–water interfaces within the emulsions. This behavior highlights the influence of internal structure and phase distribution on the dielectric response of emulsified systems.

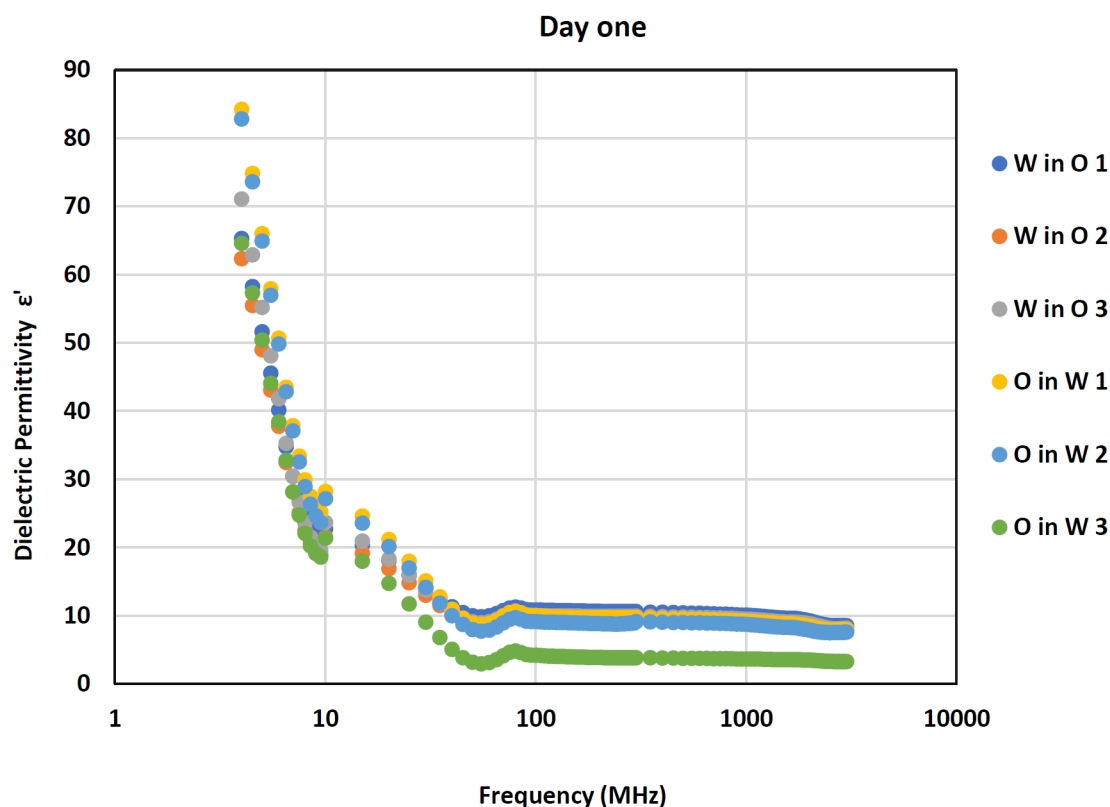


Figure 4—Permittivity profiles at different frequencies for the two emulsion types of oil in water (O/W) and water in oil (O/W) on the first day.

Moreover, the permittivity values from the first day until after two weeks were almost constant at 10 dimensionless at higher frequencies (1000) MHz and changed at the lower frequencies with a strong decline from 82 to 22 dimensionless. The visual observations show that both have the same properties by visual inspection; they are both very heavy, nevertheless, (oil in water) has become heavier than (water in oil) over time to the point that it cannot move. Oil in water is more stable than water in oil emulsion, even though they are prepared in the same proportions. After 24 hours, the permittivity in (O/W) is 10.5 dimensionless in all repeated measurements, while in (W/O) it is (6.9), (5.5), and (1.8) dimensionless during repeated measurements. All values here are taken at 1000 MHz. As for conductivity, (O/W) achieved a higher conductivity of (0.06 S/m) and (W/O) achieved (0.03 S/m) at 1000 MHz. At higher frequencies than that, specifically 2900 MHz, (O/W) achieved (0.2 S/m) and (W/O) (0.1 S/m), i.e., doubling each time. It is worth mentioning that the emulsion was prepared by mixing 50% of oil with 50% of KCl brine to produce two emulsion types (50%O / 50%W) and (50%W / 50%O). A 100 mL of oil was added to 100 mL of 3wt% KCl brine. The solution was mixed at 3000 rpm for 20 minutes to ensure a homogeneous mixture, and no surfactant was added.

The dielectric response of the emulsions on day two, as illustrated in Figure 5, is presented as a function of frequency for both water-in-oil (W/O) and oil-in-water (O/W) types. Three independent readings were taken, showing strong consistency and reinforcing measurement reliability. The recorded permittivity spans from 25 at 4 MHz to 1.7 at 3000 MHz. Compared to day one, the values are reduced, possibly due to minor shifts in droplet dispersion or emulsion structure over time. The w/o emulsion demonstrates relatively higher permittivity values, likely due to isolated water droplets contributing to interfacial polarization. A gradual decline in permittivity with increasing frequency remains evident, which is characteristic of polarization relaxation at the oil–water interface.

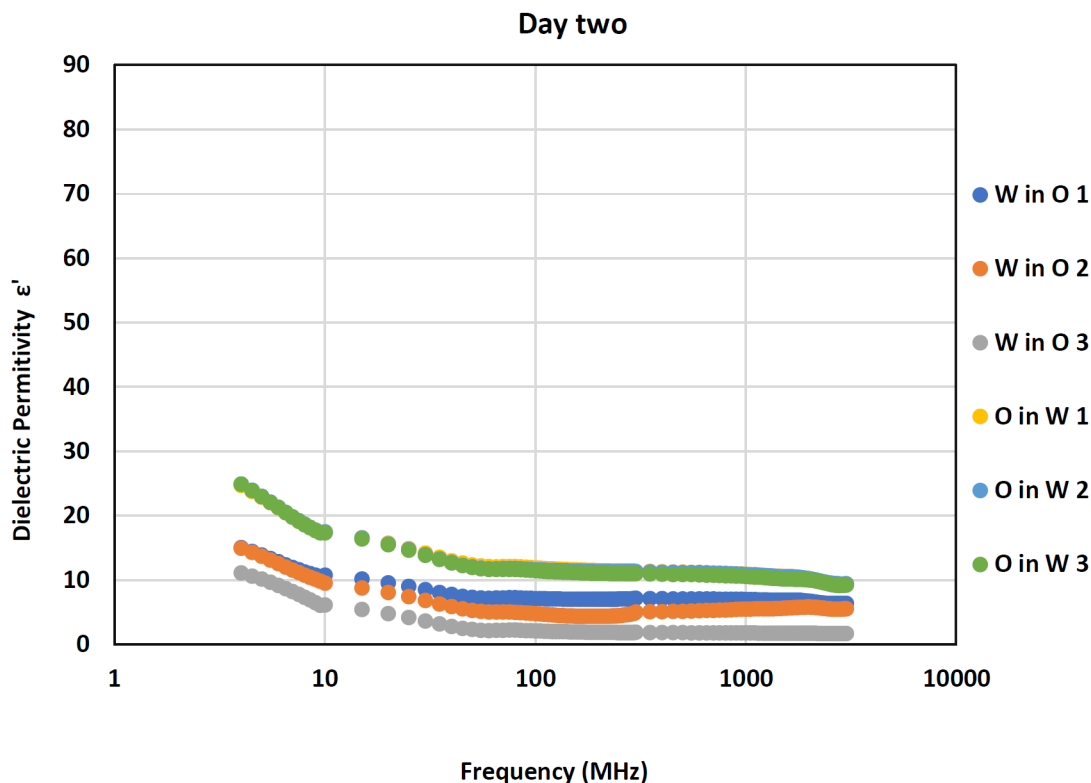


Figure 5—Permittivity dielectric behavior of emulsions on day two: frequency-dependent response and comparison between water-in-oil (W/O) and oil-in-water (O/W) systems.

In addition, the conductivity of both emulsion types—water-in-oil (W/O) and oil-in-water (O/W)—was measured as a function of frequency on day one, as illustrated in Figure 6. The results show a clear increase in conductivity with rising frequency, indicating typical dispersion behavior of heterogeneous emulsions. The measured conductivity values ranged from 0.24 S/cm at a low frequency to 0.05 S/cm at a high frequency. The W/O emulsions exhibited higher conductivity compared to the o/w systems across the measured frequency range. This observation deviates from the expected behavior, where continuous water phases in O/W emulsions typically support higher ionic mobility. The elevated conductivity in W/O samples may result from closely packed or partially connected water droplets, forming conductive pathways through the oil matrix. Additionally, differences in droplet size, salinity, or emulsion microstructure could have contributed to this unexpected trend. These results suggest the importance of internal water distribution and interfacial polarization effects, which dominate the dielectric response regardless of phase continuity.

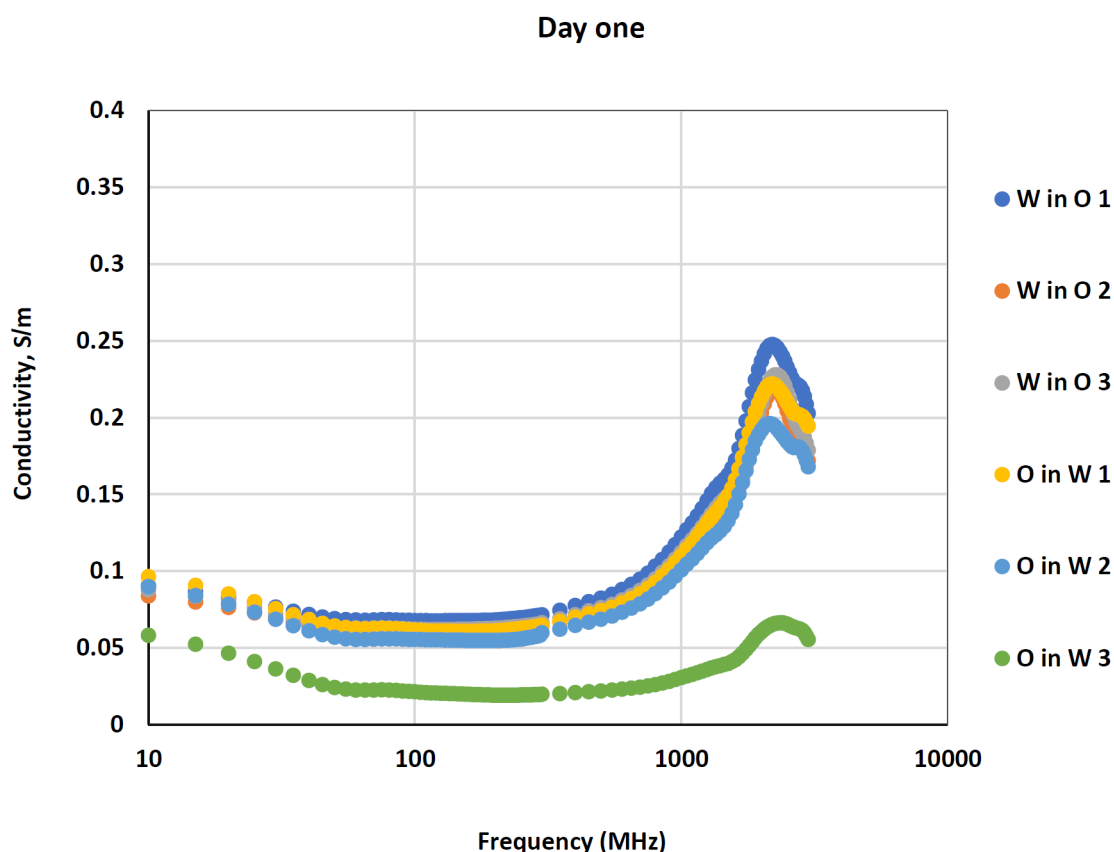


Figure 6—Frequency-dependent conductivity behavior of water-in-oil (W/O) and oil-in-water (O/W) emulsions on day one.

The conductivity profiles for water-in-oil (W/O) and oil-in-water (O/W) emulsions on day two, as shown in Figure 7, remained consistent with prior measurements, although slight variations were observed in magnitude. Conductivity values ranged from 0.23 S/cm to 0.01 S/cm over the scanned frequency interval. The O/W emulsion demonstrated higher conductivity levels, reaffirming the role of continuous aqueous pathways. In general, all samples exhibit a significant increase in conductivity at higher frequencies, particularly beyond approximately 1000 MHz, indicating enhanced ionic mobility and polarization effects at higher energy levels.

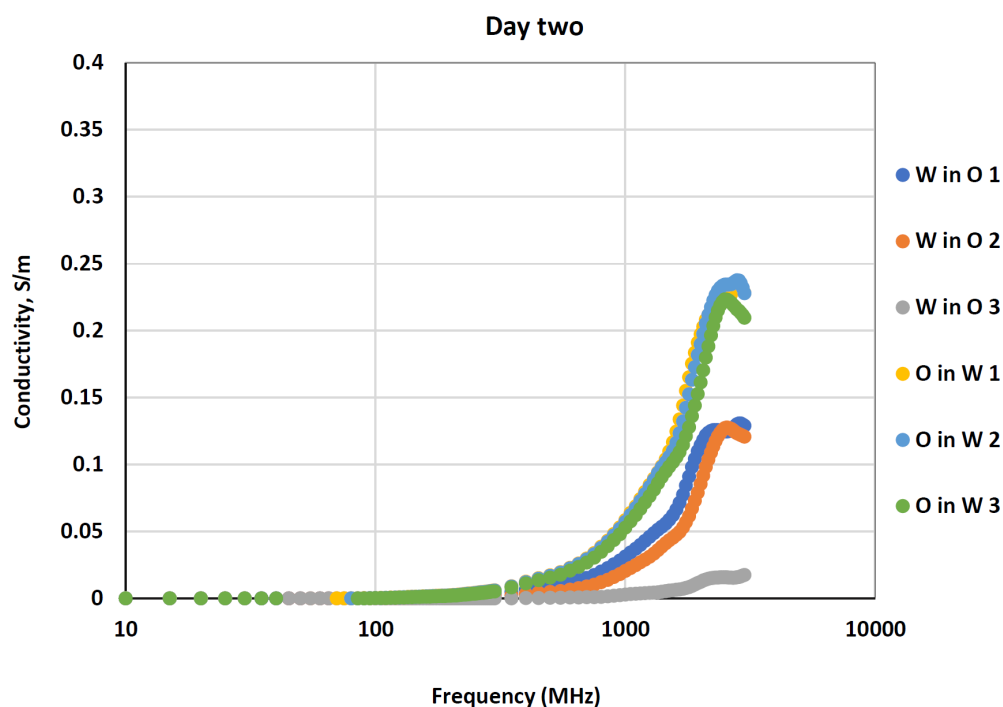


Figure 7—Comparative conductivity profiles of water in oil (W/O) and oil in water (O/W) emulsions over frequency on day two.

Microscopic Images

The visual representation in Figure 8 supports the observed emulsion instability over time, as confirmed by the experimental data. The transition from a uniform droplet distribution to noticeable sedimentation and eventual flocculation reflects the physical separation processes occurring within the system. Such behavior is commonly associated with changes in interfacial tension, droplet size growth, and lack of surfactant stabilization, all of which contribute to phase separation and reduced emulsion stability. This was illustrated in the microscopic examination in Figures 9 (a) and (b).

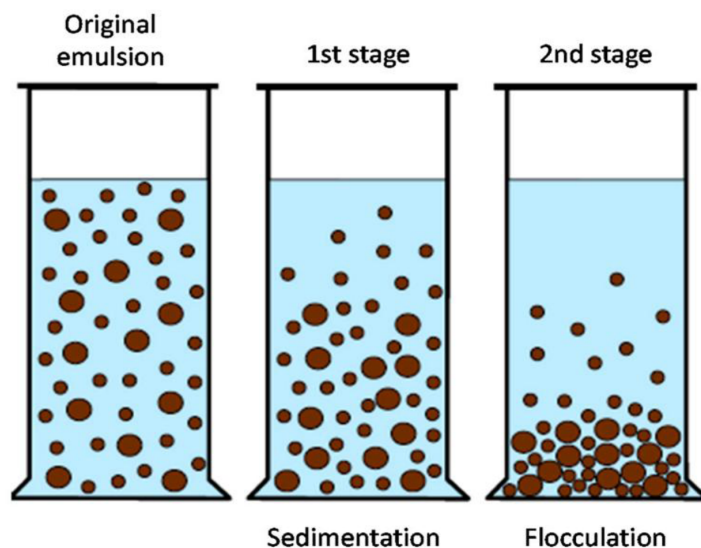


Figure 8—Progressive destabilization of oil-in-water emulsions: from uniform dispersion to sedimentation and flocculation. (Adapted from Querol et al., 2021).

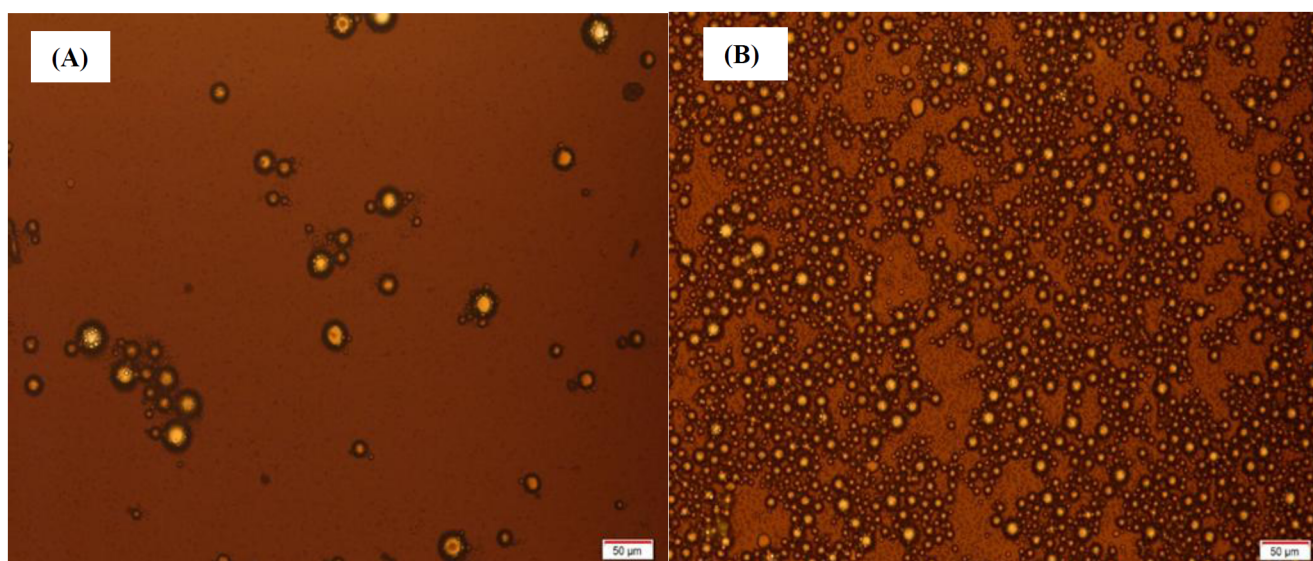


Figure 9—Microscopic images of an oil and water emulsion with a 50 μm , illustrating the dispersion of droplets within the continuous phase and providing insight into the emulsion's microstructure. A) stable emulsion with fewer larger droplets and B) an increase in droplet population and visible clustering.

Progressive destabilization of an Oil-in-Water emulsion: from uniform dispersion to sedimentation and flocculation (adapted from Querol et al., 2021). The image (Figure 8) illustrates the progressive destabilization of an oil-in-water emulsion through three distinct stages. Initially, the emulsion is well-dispersed with uniform droplet distribution. In the first stage (sedimentation), larger droplets begin to settle at the bottom due to gravitational forces. In the second stage (flocculation), droplets start to aggregate, forming clusters without fully merging. This process indicates a loss of stability over time, often influenced by factors such as droplet size, density differences, and interfacial tension.

The left image, Figure 9 (a), shows a relatively stable emulsion with fewer, larger droplets that appear well-dispersed and separated. In contrast, the right image, Figure 9 (b), reveals a significant increase in droplet population and visible clustering, indicating the onset of flocculation and reduced inter-droplet distances. These observable changes reflect the progressive breakdown of emulsion stability over time, likely influenced by factors such as droplet coalescence, gravitational separation, and increased interfacial interactions.

Investigating the impact of droplet size and viscosity on emulsion stability and dielectric behavior. The stability and dielectric behavior of emulsions are strongly influenced by both particle size and viscosity. Smaller droplet sizes and narrower distributions enhance emulsion stability by increasing surface area and promoting stronger inter-droplet interactions, such as van der Waals forces and steric hindrance (Tadros, 2013). Simultaneously, high viscosity in the continuous phase limits droplet mobility, reducing coalescence and creaming, especially in water-in-oil (W/O) emulsions (McClements, 2005). Increasing oil viscosity slows the movement of dispersed water droplets, further preventing phase separation. Viscosity also affects the dielectric response of emulsions. As viscosity increases, the mobility of dipolar molecules decreases, suppressing dielectric relaxation and shifting peaks to lower frequencies (Mahato et al., 2003). In W/O emulsions, the presence of dispersed water droplets increases both the apparent viscosity and the complexity of dielectric behavior, often resulting in interfacial polarization effects such as the Maxwell–Wagner–Sillars phenomenon (Ghosh & Sen, 1991). These effects become more pronounced with smaller droplet sizes and higher volume fractions. Recent work by Smith and Jones (2023) demonstrated a strong correlation between dielectric relaxation time and viscosity across a wide viscosity range—from less than 10 cP to over 100,000 cP—highlighting the potential of dielectric spectroscopy as a non-invasive viscosity estimation tool. Likewise, nanoemulsions with droplet sizes below 200 nm have shown superior long-

term stability due to dominant Brownian motion effects that resist gravitational separation (Solans & Solé, 2012). In a future study, more work could be conducted to explore how variations in droplet size distribution and time-dependent viscosity shifts (notably observed after Day 2) influence emulsion stability and dielectric characteristics. Special attention will be paid to how dielectric dispersion correlates with evolving rheological properties. Combining particle size analysis with broadband dielectric spectroscopy will offer a deeper understanding of emulsion dynamics, supporting more efficient design and monitoring of emulsions in industrial and petroleum applications (Jafari et al., 2020).

In summary, the primary objective of this study is to establish a clear correlation between the microstructural features of water and oil emulsions—such as droplet size, distribution, and degree of coalescence—and their corresponding dielectric behavior over a broad frequency range. By applying both optical microscopy and dielectric spectroscopy on the same set of samples, the research aims to reveal how internal emulsion architecture governs electromagnetic response. The outcomes of this work have direct implications for several oilfield applications, particularly in enhancing the interpretation of dielectric log responses, improving fluid typing accuracy, and optimizing emulsion treatment strategies. Furthermore, the methodology presented can support future developments in digital rock physics and reservoir simulation models by incorporating physically meaningful dielectric parameters that reflect real microstructural conditions. However, this study presents some limitations that should be considered when interpreting the results. First, the absence of surfactants limited the long-term stability of the emulsions, potentially affecting the consistency and repeatability of the dielectric measurements in long-term applications. Additionally, the study examined specific emulsion formulations that may not encompass the full range of behaviors exhibited under different compositional or environmental conditions, such as variations in salinity or temperature. On the other hand, the resolution of microscopic imaging was restricted to magnifications between 20 μm and 200 μm , which may not reveal finer droplet structures or nanoscale interactions. Also, the mixing procedure was fixed at 3000 rpm for 20 minutes, which did not explore the impact of different shear rates or energy inputs on emulsion structure and dielectric response.

Future studies are encouraged to incorporate surfactants or natural emulsifying agents to improve emulsion stability and assess their effect on dielectric behavior across different frequency ranges. Investigating the impact of environmental factors such as salinity, pH, and temperature could provide insights into the behavior of emulsions under more realistic reservoir and processing conditions. Expanding the study to include nanoemulsions would allow for a comparison of dielectric characteristics at different structural scales. Employing more advanced imaging techniques, such as confocal microscopy or cryogenic scanning electron microscopy (cryo-SEM), may yield more detailed observations of internal droplet morphology and phase boundaries. Furthermore, exploring the time-dependent behavior of emulsions under static and dynamic conditions could help understand the effects of coalescence, separation, and creaming on dielectric properties over time.

Conclusions

This study provided valuable insights into the dielectric behavior of water-in-oil and oil-in-water emulsions by correlating microscopic structure with electrical properties. The integrated approach enhanced understanding of emulsion stability and its implications for petroleum applications, and the following are the key findings.

- Emulsion type and composition significantly affect dielectric behavior, with clear distinctions between W/O and O/W emulsions in both permittivity and conductivity responses across the tested frequency range (4 MHz – 3 GHz).
- W/O emulsions with 50% water exhibited high initial dielectric permittivity values (up to ~ 84), which decreased notably with increasing frequency, and showed unexpectedly high conductivity (~ 0.24 S/cm), likely due to connected water droplet networks forming conductive pathways.

- O/W emulsions with 50% oil displayed slightly lower permittivity (~ 80), with a similar decreasing trend across frequency, and showed increased conductivity (~ 0.23 S/cm) on day two, aligning with the expected behavior of continuous water-phase systems.
- Over time, both emulsion types demonstrated reduced permittivity values, reflecting possible droplet redistribution or aging effects. The consistent frequency-dependent increase in conductivity suggests enhanced ionic mobility and interfacial polarization, strongly influenced by phase structure and stability.
- Microscopic imaging confirmed uniform droplet distribution in emulsions without surfactants, supporting the dielectric measurements as representing the bulk average behavior of the system.
- Mixing speed plays a critical role in emulsion stability, with instability and phase separation evident at lower mixing speeds (e.g., 990 rpm), emphasizing the need for optimized shear conditions.
- Overall, the dielectric method proves to be a reliable and non-invasive tool for distinguishing between emulsion types and stability, which can assist in diagnosing emulsion behavior and informing treatment decisions in petroleum applications.

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